LIVING WAVES: GEOLOGICAL MUSIC

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ABSTRACT

This paper describes the creative and technical processes involved in producing the multi-channel fixed-media composition *Living Waves,* commissioned for the interdisciplinary Ruskin Rocks Project[1] led by Leeds University. The project brought together musicians, artists and scientists to create two new 21st Century stone percussion instruments (the Brantwood Musical Stones[2]) to be housed at Brantwood, the former home of artist, naturalist and social commentator John Ruskin, now a museum situated in the Lake District. The two tuned percussion instruments have been created from ringing rock extracted from several Cumbrian Quarries.

Living Waves has been inspired by the thoughts, writings and paintings of Ruskin. It is the sub-title he himself gave to *Deucalion*, his book on geology. It is also the name of a recent exhibition at Brantwood of some of his drawings and watercolours in which he explored the many repeated patterns which occur in nature. Mountains are viewed as fluid and dynamic forms, living waves created by the shifting of the earth's crust.

Through the compositional process I have tried to mirror Ruskin's visual explorations from a sonic perspective, using a range of transformational techniques to convey repeated patterns in different soundscapes at both the micro and macro level. On a conceptual level, Ruskin identified with the medieval scholars' concept of *Musica Mundana*, or 'music of the spheres'. A celestial harmony. The three branches of the medieval concept of music were laid down by Boethius as *Musica Mundana, Musica Humana* (the internal music of the human body), and *Musica Instrumentalis* (sounds made by singers and instrumentalists). In the context of this piece I have interpreted these as: sounds naturally occurring in nature; speech and the sound of manmade machinery; and instrumental sounds (in this case improvisations, on the new Brantwood Musical Stones, played by percussionist Dame Evelyn Glennie).

During the period from January 2010 – September 2011 ambisonic field recordings were taken of the natural environment that the rocks are taken from as well as location recordings at several quarries including the sounds of diamond saws, water-jet cutting, and rock blasting. These contrasting soundscapes are presented as sound paintings, weaved together by more abstract musical passages employing the sounds of the Brantwood Musical Stones and passages of text by Ruskin. Sound transformation and spatialisation techniques are employed to mimic some of the patterns occurring in the sound paintings, including the use of BATBoids[3] which uses an implementation of Craig Reynolds' Boids algorithm (mimicking the flocking and swarming of birds, bees, fish etc). Close collaboration with the performer during the recording of improvised material on the instruments ensured there were adequate opportunities for transformation and mimicking between the instrumental and soundscape material.

1. REFERENCES

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Ruskin, J., *Deucalion*, D.D.Merrell, 1886, Re-produced by BiblioBazaar, 2009

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[1] http://www.leeds.ac.uk/ruskinrocks/, project lead: Prof Bruce Yardley, Dept. of Earth Sciences, Leeds University.

[2] Designed by Kia Ng (Leeds University) and Marcus de Mowbrey (freelance musical instrument maker)

[3] 'BATBoids' v 1.00 © 2007 A P Lewis, Bangor University (Wales), part of the Bangor Audio Toolkit, www.bangor.ac.uk/music/studios/BAT